

Differentially Private Hypothesis Testing

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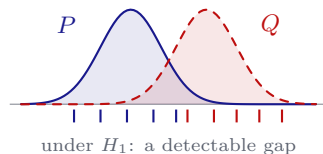
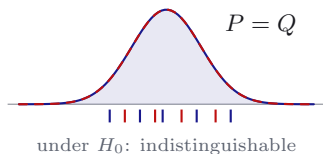
European Meeting of Statisticians, 2026

Joint work with Antonin Schrab (Cambridge)

Private Two-Sample Testing

Two samples on a **sensitive** domain: medical records, genomic data, financial data, ...

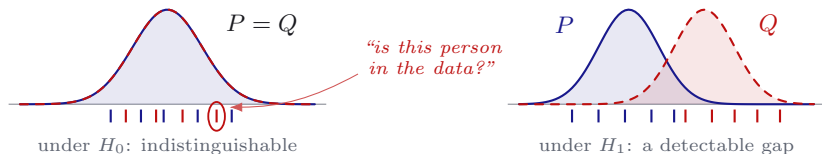
$$Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} P, \quad Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} Q, \quad H_0 : P = Q \quad \text{vs.} \quad H_1 : P \neq Q.$$



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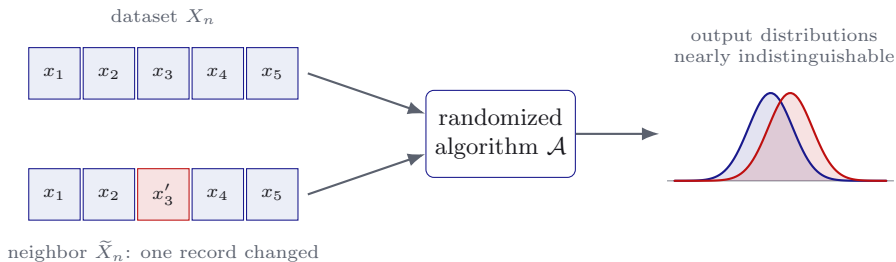
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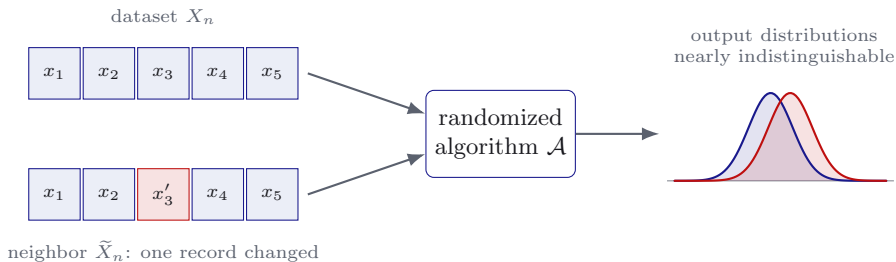


Goal: decide $P = Q$ vs. $P \neq Q$ while protecting the **privacy** of every individual.

Central Differential Privacy



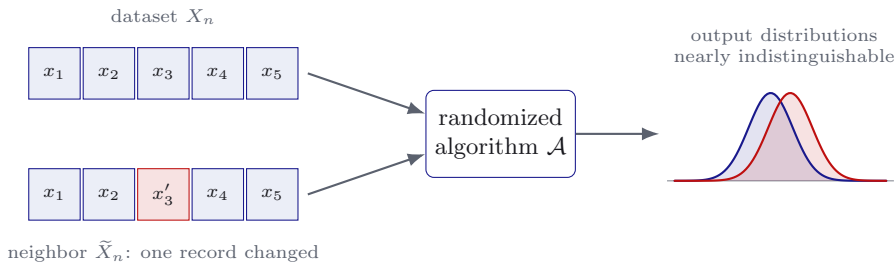
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Definition: ϵ -differential privacy

For all neighboring $X_n, \tilde{X}_n$ and all events S : $\mathbb{P}(\mathcal{A}(X_n) \in S) \leq e^\epsilon \mathbb{P}(\mathcal{A}(\tilde{X}_n) \in S)$.

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Laplace mechanism: to release a statistic T , output $T + (\Delta_T/\epsilon) \zeta$ with $\zeta \sim \text{Lap}(0, 1)$, where the sensitivity Δ_T is the largest change of T under one record.

Private Hypothesis Testing

A test is a randomized $\phi : \text{data} \rightarrow \{0, 1\}$, and *only this decision is released*.

- private level- α tests: $\Phi_{\alpha, \varepsilon} = \left\{ \phi : \sup_{P \in \mathcal{P}_0} \mathbb{E}_P[\phi] \leq \alpha \text{ and } \phi \text{ is } \varepsilon\text{-DP} \right\},$
- minimax separation: $\delta^* = \inf \left\{ \delta : \inf_{\phi \in \Phi_{\alpha, \varepsilon}} \sup_{\delta\text{-separated}} \mathbb{E}[1 - \phi] \leq \beta \right\}.$

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Challenge

- Two sources of randomness, **sampling noise** and **privacy noise**, **interact in complicated ways**.
- Carefully track both sources of uncertainty and how they interact.

Prior Work on Private Testing

Discrete distribution testing under DP is well developed in theoretical computer science.

Cai et al. (2017); Acharya–Sun–Zhang (2018, 2021); Aliakbarpour et al. (2018, 2019); Canonne et al. (2019); Zhang (2021); Canonne–Sun (2022).

Computer Science: learn from it

- ✓ sharp sample complexity
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Bridge the two communities by combining their complementary strengths.

Two Main Results

Our goals:

continuous
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finite-sample
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Part I a general recipe for private permutation testing. (Kim & Schrab, 2026)

Part II the sharp L_1 minimax boundary for smooth densities under DP. (Kim, 2026)

Part I

Differentially private permutation tests

Permutation Tests

Ingredients:

data X_n
exchangeable under H_0

statistic T
large under H_1

permutations $\pi_1, \dots, \pi_B$
i.i.d. $\text{Unif}(\Pi_n)$

$$\hat{p} = \frac{1}{B+1} \left\{ 1 + \sum_{i=1}^B \mathbf{1} \left(\underbrace{T(X_n^{\pi_i})}_{=: T_i} \geq \underbrace{T(X_n)}_{=: T_0} \right) \right\}, \quad \text{reject if } \hat{p} \leq \alpha.$$

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The question

Can we add differential privacy and keep the **exact level** for **any statistic**?

A First Naive Attempt

Privatize each statistic separately and pay by composition:

$$\widetilde{M}_i = T_i + \frac{(B+1) \Delta_T}{\varepsilon} \zeta_i \quad i = 0, \dots, B, \quad \zeta_i \stackrel{\text{iid}}{\sim} \text{Lap}(0, 1),$$

where Δ_T is the **sensitivity** of T , uniform over permutations:

$$\Delta_T := \sup_{\pi \in \Pi_n} \sup_{d_{\text{ham}}(X_n, \tilde{X}_n) \leq 1} |T(X_n^\pi) - T(\tilde{X}_n^\pi)|.$$

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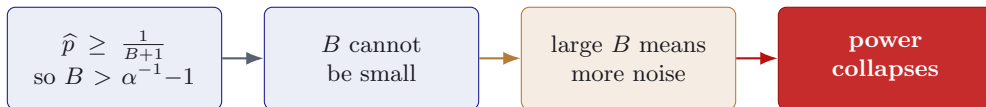
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Differentially Private Permutation Test

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$$M_i = T_i + \frac{2\Delta_T}{\varepsilon} \zeta_i \quad i = 0, 1, \dots, B, \quad \hat{p}_{\text{dp}} = \frac{1}{B+1} \left\{ 1 + \sum_{i=1}^B \mathbf{1}(M_i \geq M_0) \right\}.$$

Reject H_0 if $\hat{p}_{\text{dp}} \leq \alpha$.

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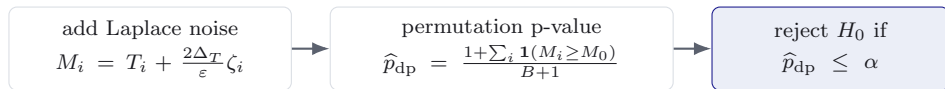
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- $M_0, \dots, M_B$ stay **exchangeable under H_0** , so $\hat{p}_{\text{dp}}$ keeps its exact level.
- **Why ε -DP?** The decision is the sign of one noisy scalar:

$$M_0 - Q_{1-\alpha^*}(M_1, \dots, M_B) = \underbrace{T_0 - Q_{1-\alpha^*}(M_1, \dots, M_B)}_{\text{sensitivity} \leq 2\Delta_T} + \underbrace{\frac{2\Delta_T}{\varepsilon} \zeta_0}_{\text{Laplace noise}}.$$

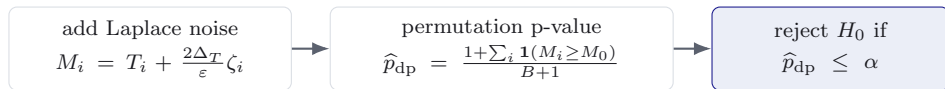
Finite-Sample Validity and Privacy

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Theorem 1 (finite-sample validity)

If X_n is exchangeable under H_0 , then for every n and every B ,

$$\sup_{P \in \mathcal{P}_0} \mathbb{P}_P(\hat{p}_{dp} \leq \alpha) = \frac{\lfloor (B+1)\alpha \rfloor}{B+1} \leq \alpha.$$

Theorem 2 (privacy)

For every P , *no exchangeability needed*,

$$\mathbf{1}\{\hat{p}_{dp} \leq \alpha\} \text{ is } \epsilon\text{-DP.}$$

Power Bound for Private Permutation Tests

Theorem 4 (uniform power: two moments + privacy)

For $B \geq 6\alpha^{-1} \log(2\beta^{-1})$, if for all $P \in \mathcal{P}_1$

$$\underbrace{\mathbb{E}_P[T(X_n)] - \mathbb{E}_{P,\pi}[T(X_n^\pi)]}_{\text{signal}} \gtrsim \underbrace{\sqrt{\frac{\text{Var}_P(T) + \text{Var}_{P,\pi}(T^\pi)}{\alpha\beta}}}_{\text{sampling noise}} + \underbrace{\frac{\Delta_T}{\varepsilon} \max\left\{\log \frac{1}{\alpha}, \log \frac{1}{\beta}\right\}}_{\text{privacy noise}},$$

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A general recipe: the mean, variance and sensitivity of T deliver power guarantees in various settings.

Private MMD Test

- The MMD (Gretton et al., 2012) is the IPM generated by the unit ball of an RKHS $\mathcal{H}_k$:

$$\text{MMD}_k(P, Q) = \sup_{\|f\|_{\mathcal{H}_k} \leq 1} \{\mathbb{E}_P[f(Y)] - \mathbb{E}_Q[f(Z)]\}.$$

- **Our test:** run the DP permutation test with the plug-in statistic $\text{MMD}_k(\hat{P}_n, \hat{Q}_n)$.

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Theorem 7 (minimum separation in the MMD metric)

For bounded kernels, the minimax separation radius under ε -DP is

$$\delta_{\text{MMD}}^* \asymp \begin{cases} n^{-1/2}, & \varepsilon \gtrsim n^{-1/2} : \text{privacy is free,} \\ (n\varepsilon)^{-1}, & \varepsilon \lesssim n^{-1/2} : \text{privacy costs.} \end{cases}$$

Part II

Smooth two-sample testing under central DP

Smooth Two-Sample Testing in L_1

Part I is optimal in the MMD metric. However, MMD depends on the kernel and **does not show the curse of dimensionality**.

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Part II: the testing problem

Given two samples of common size n from densities $f, g \in \mathcal{D}_s^d(L)$, the Hölder class of smoothness $s > 0$ and radius L on $[0, 1]^d$, test

$$H_0 : f \equiv g \quad \text{vs.} \quad H_1 : \|f - g\|_1 \geq \delta,$$

under ε -**differential privacy** constraints.

Minimax Separation Rate under Central DP

Theorem 1 (sharp L_1 minimax radius, $n\varepsilon \geq 1$)

$$\delta_{2\text{samp}}^*(n, \varepsilon) \asymp \underbrace{n^{-\frac{2s}{4s+d}}}_{\text{non-private}} \vee \underbrace{(n\sqrt{\varepsilon})^{-\frac{2s}{2s+d}} \vee (n^{3/2}\varepsilon)^{-\frac{2s}{4s+d}} \vee (n\varepsilon)^{-1}}_{\text{private}}.$$

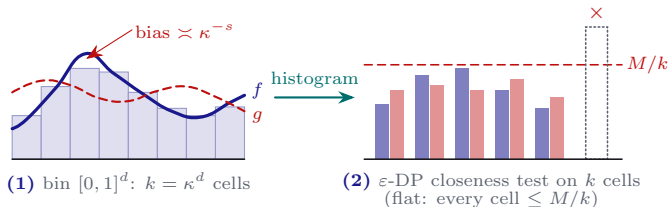
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- The first term is the **classical non-private rate**: recovered for large ε .
- The other three terms are **private rates**: dominate in different regimes for small ε .
- The second term $(n\sqrt{\varepsilon})^{-\frac{2s}{2s+d}}$ **disappears** when $s \geq d/4$.

Upper Bound via Binning and Closeness Testing



- Convert the continuous problem into a **discrete closeness testing** problem.
- Leverage **existing results** for discrete closeness testing.
- However, naively applying them to the binned data is **not optimal**.
- **Smooth and bounded densities** $\Rightarrow$ **no cell carries too much mass**; exploiting this gives the **sharp rate**.

Three Lower-Bound Methods

(1) Ingster's χ^2 method

classical term

$$\theta \in \{\pm 1\}^K, \quad P_\theta = P_0 + \theta \cdot (\text{Hölder bump}), \quad \bar{P}_\kappa := 2^{-K} \sum_{\theta} P_\theta^{\otimes n}$$
$$\chi^2(\bar{P}_\kappa, P_0^{\otimes n}) \leq e^{cn^2\kappa^{-(4s+d)}} - 1 \ll 1 \implies \text{no test detects } \bar{P}_\kappa.$$

Ingster (1987); Ingster & Suslina (2003)

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(2) DP coupling to Hamming distance

within-cell and linear terms

$$\mathbb{E}[d_{\text{ham}}(X, X')] \ll 1/\varepsilon \text{ for some coupling } X \sim Q, X' \sim P_0^{\otimes n} \implies \text{no } \varepsilon\text{-DP test detects } Q.$$

Acharya-Sun-Zhang (2018)

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Acharya–Sun–Zhang (2018)

(3) The Transport Inequality (new)

transport term

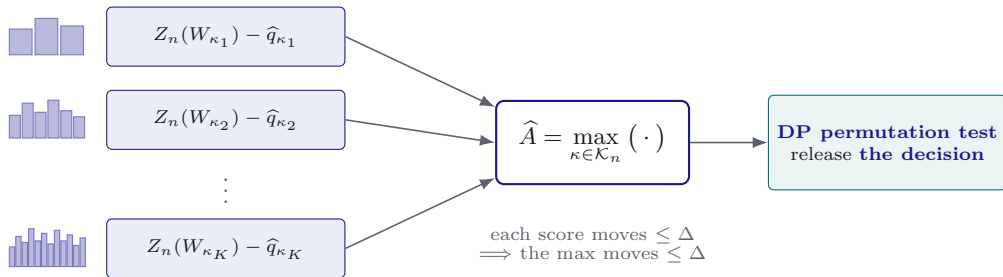
$$\text{KL}(Q \parallel P_0^{\otimes n}) \ll (n\varepsilon^2)^{-1} \implies \text{no } \varepsilon\text{-DP test detects } Q.$$

Kim (2026), arXiv:2607.23974

Adaptation

Part II, continued: unknown smoothness s

Adaptation To Unknown Smoothness



- The optimal test depends on the smoothness s through the **binning resolution**.
- When s is unknown, combine tests over a **dyadic grid of resolutions**.
- Instead of **conservative** Bonferroni correction, apply the **DP permutation test**.

Adaptive Separation Rate

Theorem 12 (adaptive upper bound)

$$\|f - g\|_1 \gtrsim \underbrace{\left(\frac{\sqrt{\log \log n}}{n} \right)^{\frac{2s}{4s+d}}}_{\text{adaptation cost } \sqrt{\log \log n}} \vee \underbrace{(n\sqrt{\varepsilon})^{-\frac{2s}{2s+d}} \vee (n^{3/2}\varepsilon)^{-\frac{2s}{4s+d}} \vee (n\varepsilon)^{-1}}_{\text{three privacy terms: do not change}}.$$

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Theorem 13 (matching adaptive lower bound)

Even for arbitrary *non-private* randomized tests, uniformly over $s \in [s_-, s_+]$,

$$\delta_{\text{adapt}}^*(n, s) \gtrsim \left(\frac{\sqrt{\log \log n}}{n} \right)^{\frac{2s}{4s+d}}.$$

Summary

1. **Part I: a general recipe for DP permutation testing.** Works for any statistic, with exact finite-sample level and ε -DP.
2. **Part II: sharp minimax separation rates under central DP.** Built on the Part I framework, for two-sample testing over Hölder classes.

Open Problems

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2. **Other metrics and classes.** Beyond L_1 and isotropic Hölder: general L_p losses, Besov, anisotropic smoothness.
3. **Other problems.** Beyond two-sample testing: regression, conditional independence, heavy-tailed models.

Thank you!



Part I

Kim & Schrab (2026)
arXiv:2310.19043



Part II

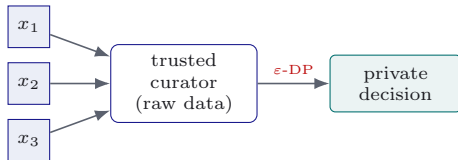
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Code: github.com/antoninschrab/dpkernel

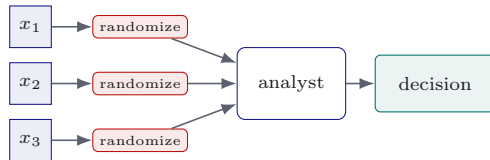
Backup slides

Backup: Central and Local Privacy Models

Central model: privatize the output



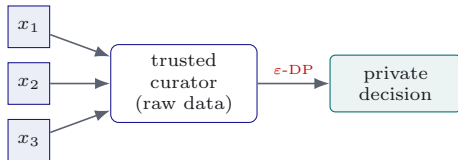
Local model: privatize before sharing



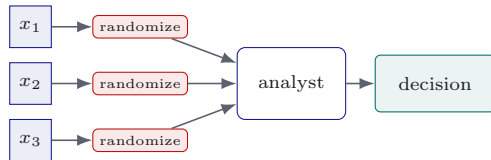
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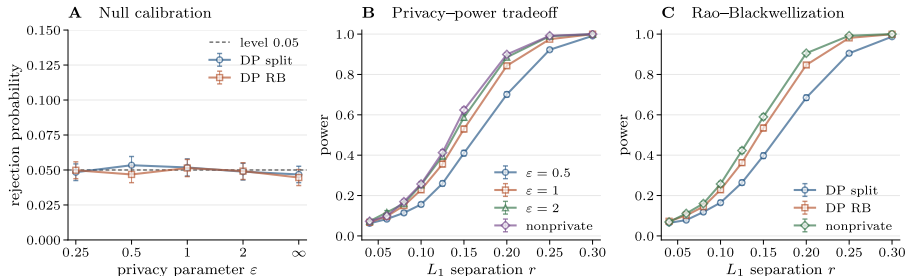
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This talk focuses on the **central model** and develops results for private hypothesis testing.

Backup: Simulation — Exact Level and Power

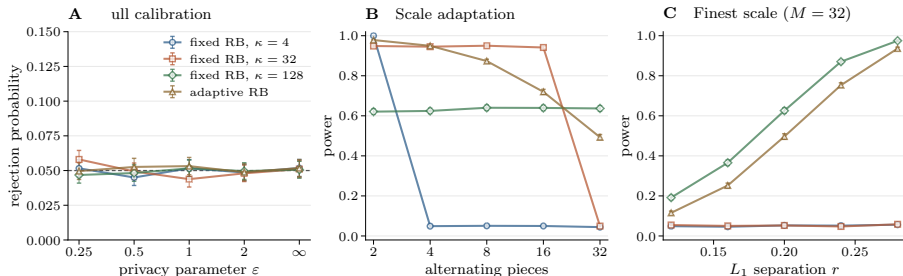


A. The test controls level exactly for every ε .

B. Privacy reduces power most in the transition region.

C. At the same privacy budget, **Rao-Blackwellization is more powerful than splitting.**

Backup: Simulation — Adaptive vs. Best Fixed Resolution



A. Level is exact for every ε , at any fixed or adaptive resolution.

B. A *coarse* κ is excellent until the alternating signs cancel within bins and it **collapses**.

C. The adaptive test **nearly matches the best fixed resolution** at each scale.

Backup: Sensitivity of the Plug-in IPM

Two samples Y_n, Z_n on $\mathcal{S}$, $T = \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum f(Y_i) - \frac{1}{n} \sum f(Z_i) \right|$.

$$\Delta_T = \frac{1}{n} \sup_{X, X' \in \mathcal{S}} \sup_{f \in \mathcal{F}} |f(X) - f(X')| \quad (\text{exactly}).$$

- Mean difference in ℓ_p : $\sup \|X - X'\|_p$.
- Total variation: ≤ 2 . Kolmogorov: ≤ 1 .
- Wasserstein-1: $\leq \sup \|X - X'\|$.
- MMD: $\leq \sqrt{2K}$ for $0 \leq k \leq K$.

Theorem 4 then gives power $\geq 1 - \beta$ whenever

$$\text{IPM}_{\mathcal{F}}(P, Q) \gtrsim \frac{R_n(\mathcal{F})}{\sqrt{\alpha\beta}} + \frac{\sqrt{n} \Delta_T}{\sqrt{\alpha\beta}} + \frac{\Delta_T}{\varepsilon} \max \left\{ \log \frac{1}{\alpha}, \log \frac{1}{\beta} \right\},$$

with $R_n(\mathcal{F})$ the worst-case permuted Rademacher complexity.

Backup: Discrete Closeness Testing Rates

Unrestricted simplex (minimax optimal; matches Zhang, 2021)

$$n \gtrsim \frac{\sqrt{k}}{\tau^2} + \frac{k^{2/3}}{\tau^{4/3}} + \frac{\sqrt{k}}{\tau\sqrt{\varepsilon}} + \frac{k^{1/3}}{\tau^{4/3}\varepsilon^{2/3}} + \frac{1}{\tau\varepsilon}$$

Bounded-histogram subclass $\mathcal{P}_{k,M}$

$$n \gtrsim \frac{\sqrt{k}}{\tau^2} + \frac{\sqrt{k}}{\tau\sqrt{\varepsilon}} + \frac{k^{1/3}}{\tau^{4/3}\varepsilon^{2/3}} + \frac{1}{\tau\varepsilon}$$

Moment bounds behind this, with $m = \lfloor n/2 \rfloor$:

$$\mathbb{E}[\bar{Z}^{(0)}] \gtrsim \min \left\{ m\tau, \frac{m^2\tau^2}{k}, \frac{m^{3/2}\tau^2}{\sqrt{k}} \right\}, \quad \text{Var}(\bar{Z}^{(0)}) + \text{Var}(\bar{Z}^\pi) \lesssim \min \left\{ \frac{m^2}{k}, m \right\}.$$

Both $B_{\text{perm}} = \lceil 6\gamma^{-1} \log(2\beta^{-1}) \rceil$; type I error $\leq \gamma$ exactly.

Backup: Binning Lemma and Flat Histograms

Binning lemma: signal survives

For $h = f - g$ and cells C_j of volume $1/k$,

$$p_f^{(\kappa)}(j) = \int_{C_j} f,$$

$$\|p_f^{(\kappa)} - p_g^{(\kappa)}\|_1 = \|W_{\kappa,d} h\|_1 \geq \frac{1}{2} \|h\|_1 - C\kappa^{-s}.$$

On $O(\kappa^{-1})$ blocks, h is $O(\kappa^{-s})$ -close to a Taylor polynomial; cell averaging preserves a fixed fraction of its L_1 norm.

Structural gain: flat histograms

Since $\|f\|_\infty, \|g\|_\infty \leq M$,

$$p_f^{(\kappa)}(j) \leq M|C_j| = M\kappa^{-d} = \frac{M}{k},$$

$$p_f, p_g \in \mathcal{P}_{k,M} := \{v \in \Delta_k : \|v\|_\infty \leq M/k\}.$$

Thus no cell can carry more than a constant multiple of the uniform mass $1/k$.

Why flatness matters for the discrete test:

$$p, q \in \mathcal{P}_{k,M} \implies \text{Var}(Z) + \text{Var}(Z^\pi) \lesssim_M \min\left\{\frac{n^2}{k}, n\right\} \implies \frac{k^{2/3}}{1^{4/3}}.$$

Cell masses $O(1/k)$ imply $O_M(n^2/k)$ same-cell pairs, which control the variance; the privacy terms of the discrete rate are unchanged.

Backup: The Split Statistic

Halve each sample, bin into k cells, and compare the four count vectors (Diakonikolas et al., 2021):

$$Z_n = \sum_{i=1}^k \left\{ \underbrace{|C_i^{X,1} - C_i^{Y,1}| + |C_i^{X,2} - C_i^{Y,2}|}_{\text{cross-group}} - \underbrace{|C_i^{X,1} - C_i^{X,2}| + |C_i^{Y,1} - C_i^{Y,2}|}_{\text{within-group}} \right\}$$

$C_i^{X,j}$: number of group- X points in split-half j falling in bin i (similarly $C_i^{Y,j}$).

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- **Mean zero under H_0** ; only the **cross** terms grow under H_1 .

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- **Mean zero under H_0** ; only the **cross** terms grow under H_1 .
- **Sensitivity ≤ 4** : constant and explicit.

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$C_i^{X,j}$: number of group- X points in split-half j falling in bin i (similarly $C_i^{Y,j}$).

- **Mean zero under H_0** ; only the **cross** terms grow under H_1 .
- **Sensitivity ≤ 4** : constant and explicit.
- **DP permutation test applies**: **exact level**, ϵ -DP, **no unspecified constants**.

Backup: Rao–Blackwellization of the Split Statistic

The split statistic **uses only half the data** and depends on an **artificial random split**; the remedy is to average the split out, conditionally on everything observed,

$$\overline{Z} = \mathbb{E}_\lambda[Z_m(W, \lambda) \mid W, \ell], \quad m = \lfloor n/2 \rfloor.$$

Preserved	same mean, smaller variance; sensitivity still ≤ 4 (uniform in λ), so privacy and calibration carry over verbatim
Gained	uses <i>all</i> observations; handles unequal sample sizes; exact $O(n)$ evaluation, despite exponentially many splits

Backup: Lower Bound Constructions

barrier	rate	mechanism
classical	$n^{-\frac{2s}{4s+d}}$	random-sign hypercube of Hölder bumps + χ^2 second moment; no privacy used
within-cell	$(n\sqrt{\varepsilon})^{-\frac{2s}{2s+d}}$	$\pm$ bumps cancel <i>inside each cell</i> : the histogram sees nothing ; DP coupling forces Hamming cost $\gtrsim 1/\varepsilon$
transport	$(n^{3/2}\varepsilon)^{-\frac{2s}{4s+d}}$	new inequality $\mathbb{E}_Q[p_M] - \mathbb{E}_0[p_M] \lesssim \varepsilon\sqrt{n \text{KL}(Q\ P_0^n)}$: a private test needs $\text{KL} \gtrsim (n\varepsilon^2)^{-1}$ to move
linear	$(n\varepsilon)^{-1}$	fixed smooth perturbation: expected Hamming $\asymp n\ f - 1\ _1$, then DP coupling

Backup: Phase Transition Points

Write $\varepsilon = n^{-\alpha}$ and ask which term of Theorem 1 dominates as α grows:

$s < d/4$: four phases

$$\alpha_{12} = \frac{4s}{4s+d}, \quad \alpha_{23} = \frac{d-2s}{d}, \quad \alpha_{34} = \frac{s+d}{2s+d}.$$

Example $s = 1, d = 8$: $\frac{1}{3}, \frac{3}{4}, \frac{9}{10}$.

$s \geq d/4$: three phases

the second term never dominates;
transitions at

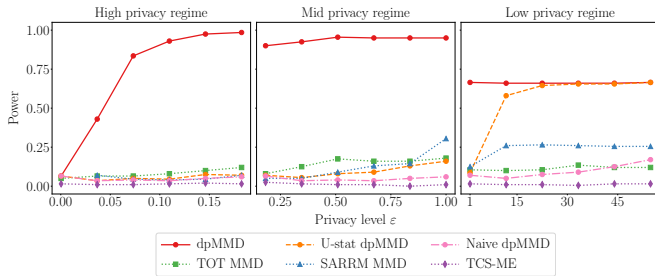
$$\alpha = \frac{1}{2} \quad \text{and} \quad \alpha_{34} = \frac{s+d}{2s+d}.$$

Example $s = 2, d = 4$: $\frac{1}{2}, \frac{3}{4}$.

Optimal resolutions in each regime:

$$\kappa_1 \asymp n^{\frac{2}{4s+d}}, \quad \kappa_2 \asymp (n\sqrt{\varepsilon})^{\frac{2}{2s+d}}, \quad \kappa_3 \asymp (n^{3/2}\varepsilon)^{\frac{2}{4s+d}}.$$

Backup: dpMMD Power in Practice



The U-statistic-based private test is essentially powerless in the high- and mid-privacy regimes and only recovers as ϵ grows: its sensitivity is larger than the plug-in's. The naive (B -dependent) construction degrades as B grows.

Backup: Comparison with Subsample-and-Aggregate

- Subsample-and-aggregate (Nissim et al., 2007) converts *any* non-private test into a private one, fully black-box.
- Price of that generality:
 - ▶ generally **suboptimal power**; typically fails to recover the optimal sample complexity;
 - ▶ performance is sensitive to the number of subsamples, whose optimal choice is open.
- Our route is the opposite trade: we use the structure of the statistic (its sensitivity) and keep optimality.

Also in the appendix of Part I

Rényi-DP variant via the Gaussian mechanism, retaining finite-sample validity, privacy, consistency, and a general uniform power guarantee.

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